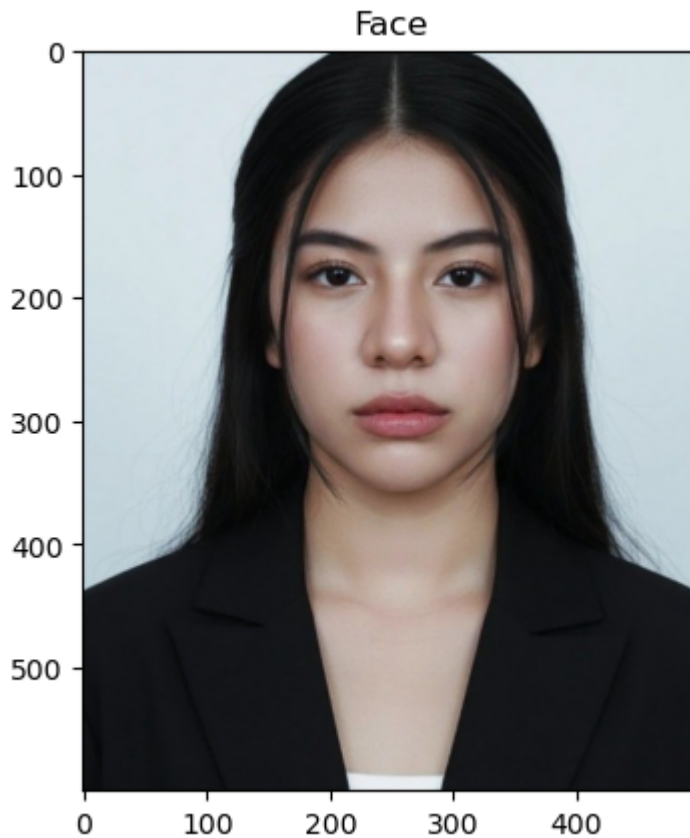


```
In [1]: # Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
In [3]: # Load the Face Image
faceImage = cv2.imread('pic1.jpeg')
plt.imshow(faceImage[:, :, ::-1]);plt.title("Face")
```

```
Out[3]: Text(0.5, 1.0, 'Face')
```

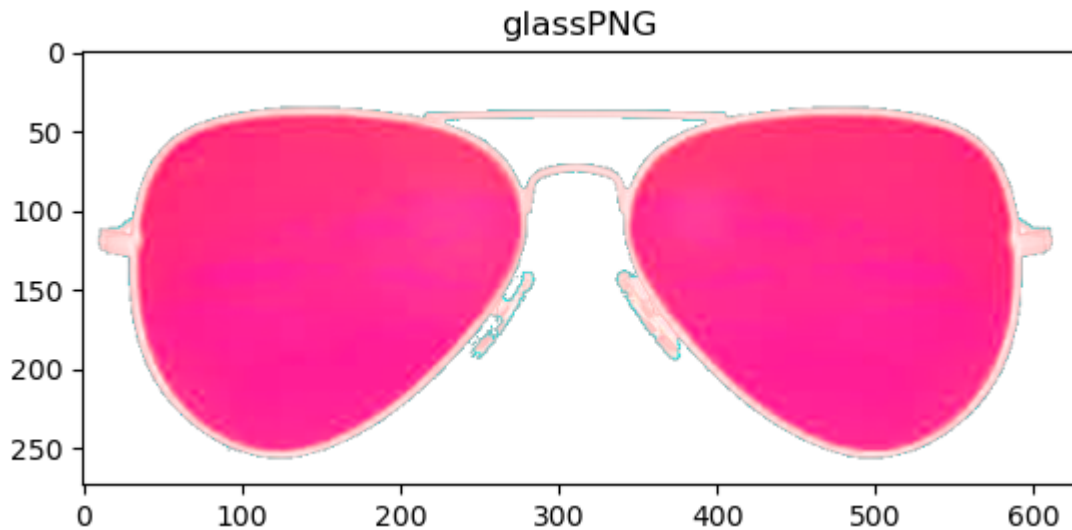


```
In [4]: faceImage.shape
```

```
Out[4]: (600, 500, 3)
```

```
In [5]: glassPNG = cv2.imread('sun (2).png',-1)
plt.imshow(glassPNG[:, :, ::-1]);plt.title("glassPNG")
```

```
Out[5]: Text(0.5, 1.0, 'glassPNG')
```

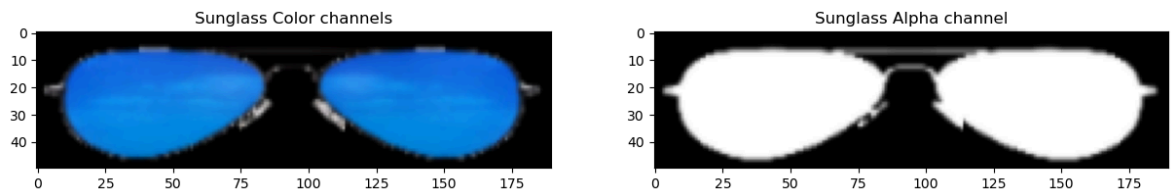


```
In [30]: glassPNG = cv2.resize(glassPNG,(190,50))
print("image Dimension ={}".format(glassPNG.shape))
```

image Dimension =(50, 190, 4)

```
In [31]: # Separate the Color and alpha channels
glassBGR = glassPNG[:,:,:3]
glassMask1 = glassPNG[:,:,:3]
```

```
In [32]: # Display the images for clarity
plt.figure(figsize=[15,15])
plt.subplot(121);plt.imshow(glassBGR[:,:,:-1]);plt.title('Sunglass Color channe
plt.subplot(122);plt.imshow(glassMask1,cmap='gray');plt.title('Sunglass Alpha ch
```

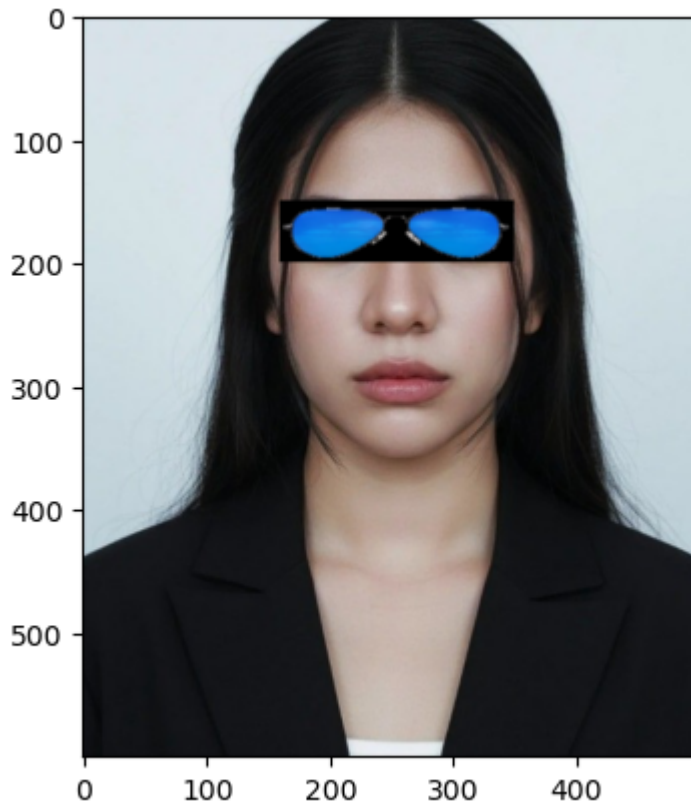


```
In [33]: # Make a copy
#faceWithGlassesNaive = resized_faceImage.copy()
faceWithGlassesNaive = faceImage.copy()

# Replace the eye region with the sunglass image
faceWithGlassesNaive[150:200,160:350]=glassBGR

plt.imshow(faceWithGlassesNaive[...,:-1])
```

Out[33]: <matplotlib.image.AxesImage at 0x13e217ea7d0>



```
In [35]: # Make the dimensions of the mask same as the input image.
# Since Face Image is a 3-channel image, we create a 3 channel image for the mas
glassMask = cv2.merge((glassMask1,glassMask1,glassMask1))

# Make the values [0,1] since we are using arithmetic operations
glassMask = np.uint8(glassMask/255)

# Make a copy
faceWithGlassesArithmetic = faceImage.copy()

# Get the eye region from the face image
eyeROI= faceWithGlassesArithmetic[150:200,160:350]

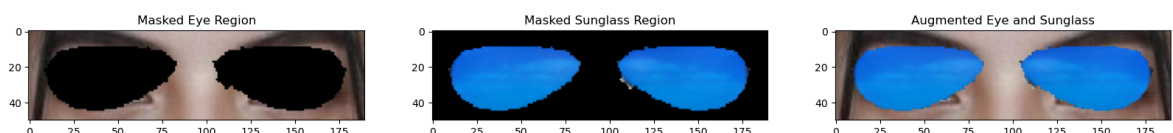
# Use the mask to create the masked eye region
maskedEye = cv2.multiply(eyeROI,(1- glassMask ))

# Use the mask to create the masked sunglass region
maskedGlass = cv2.multiply(glassBGR,glassMask)

# Combine the Sunglass in the Eye Region to get the augmented image
eyeRoiFinal = cv2.add(maskedEye, maskedGlass)

# Display the intermediate results
plt.figure(figsize=[20,20])
plt.subplot(131);plt.imshow(maskedEye[...,:-1]);plt.title("Masked Eye Region")
plt.subplot(132);plt.imshow(maskedGlass[...,:-1]);plt.title("Masked Sunglass Re
plt.subplot(133);plt.imshow(eyeRoiFinal[...,:-1]);plt.title("Augmented Eye and
```

Out[35]: Text(0.5, 1.0, 'Augmented Eye and Sunglass')



In []:

```
In [36]: # Replace the eye ROI with the output from the previous section
faceWithGlassesArithmetic[150:200,160:350]=eyeRoiFinal

# Display the final result
plt.figure(figsize=[20,20]);
plt.subplot(121);plt.imshow(faceImage[:,:,:-1]); plt.title("Original Image");
plt.subplot(122);plt.imshow(faceWithGlassesArithmetic[:,:,:-1]);plt.title("With
```



In []:

In []: